JMAN GROUP

CRM Churn Prediction Pipeline

Project Documentation

Project Overview

This project builds a machine learning pipeline to predict contractor churn at the point of renewal for a B2B contractor accreditation CRM. The model scores each contractor before their renewal date so the team can intervene proactively.

Objective

Predict whether a contractor will churn (not renew) or be retained in the upcoming renewal cycle. The model operates on pre-renewal data only, no post-outcome information is used.

Target Variable

Variable	Value	Meaning
churn_label	1	Churned - contractor did not renew
churn_label	0	Won - contractor renewed

Primary Metric

PR-AUC (Precision-Recall AUC) is used as the primary evaluation metric. It is robust to class imbalance and focuses on the minority churn class, unlike ROC-AUC which can be misleadingly high when the majority class dominates.

Data Sources

Source	Description	Key Signal
Billings	One row per renewal cycle - prices, bands, tenure, audit and membership status	Core renewal features + churn label
Renewal Calls	Call centre recordings of renewal conversations	Churn intent, complaints, price discussions
CC Calls	General contact centre calls in the 14-day pre-renewal window	Sentiment, platform issues, financial hardship
Emails / CRM	CRM email notes from the pre-renewal period	Accreditation progress, payment intention, satisfaction

Dataset File Locations

File	Path	Description
billings_preprocessed.csv	dataset/preprocessed/	Cleaned billings data
renewal_calls_preprocessed.csv	dataset/preprocessed/	Cleaned renewal call data
cc_calls_preprocessed.csv	dataset/preprocessed/	Cleaned CC call data
emails_preprocessed.csv	dataset/preprocessed/	Cleaned email/CRM data
final_merged.csv	dataset/preprocessed/	All four sources merged
df_model.csv	dataset/feature_store/	EDA-cleaned, feature-selected dataset
final_pre-renewal_churn_dataset.csv	dataset/feature_store/	Final modelling-ready dataset with selected features + label

2. Pipeline Overview

The pipeline consists of five stages executed in sequence:

Stage	Notebook(s)	Input	Output
1. Preprocessing	billings_cleaning, cc_calls_cleaning, emails_cleaning, renewal_calls_cleaning	Raw CSV files	dataset/preprocessed/ (4 files)
2. Merge	merge.ipynb	4 preprocessed CSVs	dataset/preprocessed/final_merged.csv
3. EDA	analysis.ipynb	final_merged.csv	dataset/feature_store/df_model.csv
4. Feature Selection	FeatureSelection.ipynb	df_model.csv	dataset/feature_store/final_pre-renewal_churn_dataset.csv
5. Model Building	hptuning_mbuilding.ipynb	final_pre-renewal_churn_dataset.csv	Trained model + evaluation charts

3. Data Preprocessing

Each of the four raw data sources was cleaned independently before merging. All preprocessing followed the principle of using domain knowledge to fill nulls meaningfully rather than applying generic imputation.

3.1 Billings

Unit of analysis: one row = one renewal cycle. Unique key: Co_Ref + Prospect_Renewal_Date.

Columns Dropped

Column	Reason
Last_Years_Date_Paid	100% null
Last_Payment_Score / Last_Discount_Score	Always 0 - zero variance
Billings_Contractor	Always 1 - constant flag
DateTime_Out	Exact duplicate of Renewal_Month
Current_Anchor_List	100% null

Data Type Fixes

Fix	Columns
DD-MM-YYYY strings → datetime64	Renewal_Month, Proforma_Date, Registration_Date, Prospect_Renewal_Date, Closed_Date, Last_Renewal
True/False booleans → 1/0 int	Proforma_Auto_Renewal, Proforma_World_Pay_Token
y/n strings → 1/0 int	Current_Auto_Renewal_Flag, Current_World_Pay_Token
Casing normalisation (.str.title())	Proforma_Account_Stage - unified 'vetting' and 'Vetting'
Whitespace strip	Band, Last_Band, Tenure_Group, Connection_Group, Anchor_Group

Data Error Correction

Error	Fix
Renewal_Year = 2050 (data entry error)	Replaced with year extracted from Prospect_Renewal_Date
Tenure_Years missing where Registration_Date exists	Recomputed as (Prospect_Renewal_Date − Registration_Date).days / 365.25

Churn Label

churn_label = 1 (Churned), 0 (Won).

Rows with Prospect_Outcome = 'Open' were excluded.

Six noise statuses were dropped before label creation: Duplicate Entry, Do Not Recognise Client, Existing Safecontractor Member (+ anchored variant), Existing Safemark Member, Campaign on Hold.

Null Handling

Column(s)	Fill Value	Rationale
Connection_Net, Connection_Qty, Starting_Connection_*	0	94% missing - no connections present
Discount_Amount	0	90% missing - no discount applied
Last_Years_Price	Current Amount	Missing = new customer (first renewal); is_new_customer flag added
Payment_Timeframe	0 + missing flag column	Missing = unknown timing
Last_Band	'New_Customer'	Missing = new customer history
Last_Connections, Last_Total_Net_Paid	0	No prior history
Proforma_Auto_Renewal, Proforma_World_Pay_Token	Current equivalent column	~15% missing; current flag is a reliable proxy

Grouping / Encoding

Prospect Status: 31 raw values → 10 groups (Retained, Retained_Pending, At_Risk_Engaging, At_Risk_Disengaging, Churned_Price, Churned_Business, Churned_Competitor, Churned_Service, Churned_Accreditation, Churned_Disengaged).

Audit Status: 31 raw values → 7 groups (Accredited, Renewal_In_Progress, Renewal_Failed, Initial_In_Progress, Initial_Failed, Suspended_Failed, Withdrawn).

3.2 CC Calls

Only calls within the 14-day pre-renewal window were retained (0 ≤ days_to_renewal ≤ 14).

Task	Detail
Time window filter	0 ≤ (Prospect_Renewal_Date − Call_Date) ≤ 14 days
cc_care_package	Non-standard values mapped to Standard / Express / Premier / Not Discussed; NaN → Not Discussed
13 Yes/No flag columns	Standardised to Yes / No / Not Mentioned (NaN and placeholders → Not Mentioned)
cc_agent_cross_sell_attempt	Unresolved values → Yes (implicit cross-sell)
cc_call_initiated_by	Standardised to Customer / Agent / Not Relevant
Sentiment (cc_contractor_sentiment)	Non-standard values imputed from score average: ≥70 → Satisfied, 40-69 → Neutral, <40 → Dissatisfied
Score columns (start/end/overall/issues)	Out-of-range values imputed from cleaned sentiment label (Satisfied=80, Neutral=50, Dissatisfied=20)
5 pricing/complaint flags	Any value not explicitly Yes → No
Final dtypes	Boolean columns → bool; categorical columns → pandas Categorical

3.3 Emails / CRM

Filtered to Time_to_Renewal == 'pre_renewal' only.

Task	Detail
Null groups	Three tiers (~3,155 / ~1,518 / ~1,630 nulls) from different pipeline batches; all imputed with 'Not Discussed'
normalise_flag() function	Priority rules: NaN/placeholder → None; 'Not applicable' → Not Discussed; exact match → keep; starts-with match → truncate; keyword scan; fallback → Not Discussed
19 binary/ternary flag columns	Normalised then remaining NaN → Not Discussed
crm_contractor_sentiment	Mapped: mislabelled Yes/No fixed; free-text containing 'dissatisfied'/'negative' → Dissatisfied; NaN → Not Discussed
crm_contractor_sentiment_score	Text labels converted to numbers (Dissatisfied=20, Neutral=50); out-of-range → NaN; NaN → column median
crm_agent_chase_count	Vague words mapped (A few=3, Several=4, Multiple=5, Many=6); digits extracted from free text; NaN → 0
crm_auto_renewal_status	Valid: 0/1/2; free text and out-of-range → 0; NaN → 0

crm_membership_level	~18 variants → 5 canonical levels: Premier, Standard, Accredited, In Progress, Not Discussed

3.4 Renewal Calls

Filtered to the 14-day pre-renewal window. Rows where Co_Ref or Analysed_Call is null were dropped.

Task	Detail
Call_Direction	OUT_BOUND → Outbound; IN_BOUND → Inbound
15 Yes/No flag columns	8-step pipeline: strip formatting → handle Yes/No ambiguity → Not Applicable → Not Discussed → No → Yes → fill NaN → mark remaining as Ambiguous
Desire_To_Cancel	Free text → 6 categories via keyword matching: Desired to Cancel, Renewed, Undecided, Not Discussed, Not Applicable, Ambiguous
Explicit_Competitor_Mention, Mentioned_Competitors	Noise removed; starts-with 'yes' → Yes; else → No
Competitor_Value_Comparison	Keywords → Better Value / Similar Value / No Comparison
Competitor_Benefits_Mentioned	Keywords → Price / Service / Offering / Ease / None
Topic_Introduced_By / Customer_Response	Keyword classification into Agent/Customer/Not Discussed and Positive/Negative/Neutral/Not Discussed
10 category columns (Churn_Category etc.)	Strip whitespace; underscores → spaces; nan/N/A/Not Applicable → Not Mentioned

3.5 Merge

A year_key (Co_Ref + '_' + year) was created on all four tables as the join key. Interaction tables were aggregated per year_key before joining to Billings (left join).

Aggregation Strategy

Signal Type	Aggregation	Rationale
Binary flags (Yes/No)	max	1 if ANY interaction had the signal
Call/email counts	count	Total interaction volume = friction proxy
Numeric scores	mean, min, last_value	Trend + worst case + most recent
High-cardinality categoricals	Binary explode (one col per unique value)	Preserve all signals; 1 = appeared at least once
Low-cardinality categoricals	meaningful_mode (ignore non-signal values)	Most common real category

Post-Merge Null Filling

Column Type	Fill Value
Count columns (rc_total_calls, cc_total_calls, em_total_emails)	0 - no join match = no interactions

All binary/numeric interaction columns (rc_*, cc_*, crm_*, em_*)	0 - no interaction = no signal
All categorical interaction columns	'No_Interaction' - explicit label for absence of contact
High-null columns (days_to_renewal_mean/min/last, Direction)	Dropped after merge

4. Exploratory Data Analysis

Input: final_merged.csv.

Output: dataset/feature_store/df_model.csv

4.1 Leakage Detection & Column Drops

Closed_Date was identified as a post-outcome timestamp - a gap analysis (Closed_Date − Prospect_Renewal_Date) showed the gap distribution differed sharply between Won and Churned, proving it encodes the outcome. It was dropped along with all other date columns already encoded as derived features.

Dropped	Reason
Closed_Date	Post-outcome timestamp - leakage
Prospect_Renewal_Date, Registration_Date, Proforma_Date, Renewal_Month	Already encoded as Tenure_Years and proforma_delay
Co_Ref	Identifier only
Prospect_Status / Prospect_Status_Group	Label proxy - encodes the outcome journey
Total_Net_Paid	Payment = outcome itself - leakage
Last_Total_Net_Paid	Previous year payment - leakage risk

4.2 Outlier Handling

An IQR-based outlier report was run on all continuous columns. Two columns contained genuine data entry errors and were removed by row-level filtering:

Column	Rule	Action
Payment_Timeframe	< −500 days	Rows removed
proforma_delay	> 500 days	Rows removed

All remaining outliers were left untreated. Tree-based models are robust to outliers, and the extreme values (e.g., high billing amounts, large connection counts) represent valid business scenarios. crm_agent_chased_contractor was dropped as redundant with crm_agent_chase_count.

4.3 Near-Zero Variance - Lift-Based Retention

Rather than dropping all near-zero variance columns, a lift analysis was applied:

```
lift = churn_rate_when_rare_event / overall_churn_rate
```

Columns with lift > 2 were kept despite low frequency. Columns with lift ≤ 2 were dropped. The 11 retained rare-event columns include: rc_Explicit_Switching_Intent, cc_contractor_suggest_leave, crm_competitors_mentioned, rc_Serious_Complaint, rc_Price_Switching_Mentioned,

crm_platform_issues_raised, crm_refund_mentioned, cc_business_struggles_financial_hardship, and others.

4.4 Correlation & Redundancy Removal

A billing column correlation heatmap identified highly redundant pairs. The following were dropped, keeping the most interpretable current-year equivalent:

Dropped	Kept	Correlation
Starting_Net, Starting_Vat, Starting_Gross	Gross	1.00 / mathematically derived
Starting_Membership_Net	Membership_Net	0.98
Starting_PQQ_Net	PQQNet	0.91
Connection_Net, Starting_Connection_*	Connection_Qty	> 0.90
Membership_Net (vs Gross)	Gross	> 0.85
Status_Scores	Total_Renewal_Score_New	Correlated
Proforma_Approved_Lists	Current_Anchorings	Correlated
em_total_emails	crm_contractor_sentiment_score	Correlated count vs signal

Two price features were engineered:

```
price_change       = Gross - Last_Years_Price
price_increase_pct = (price_change / Last_Years_Price) × 100
```

4.5 Interaction Column Encoding (Tri-State)

RC/CC/EM columns after merging had three states: No_Interaction (no call occurred), False (call occurred, signal absent), True (call occurred, signal present). These were split into two binary columns:

```
{col}_interacted  → 1 if any interaction occurred
{col}_true        → 1 if the signal fired
```

Individual _interacted flags were then collapsed into three master flags (had_rc_interaction, had_cc_interaction, had_em_interaction) to avoid column bloat.

4.6 Key EDA Findings

Feature	Finding
Band	Band A (top tier): ~22% churn → Band J (bottom): ~5% churn - inverse relationship
Tenure_Group	Year 1: ~21% churn → Year 4+: ~7% churn - longer tenure = lower churn
Proforma_Membership_Status	Member Only: ~29% churn vs Accredited: ~6% churn
Audit_Status_Group	Initial_Failed: ~34% churn vs Accredited: ~6% churn
Payment_Method	BACS: ~0% churn; UNKNOWN: ~100% churn - strong payment signal
price_change	Churned customers experienced larger price increases
rc_total_calls	More renewal calls correlated with higher churn (calls = escalation proxy)
Current_Anchorings	Retained customers had more active anchor relationships

4.7 Hypothesis Testing

Test	Applied To	Criteria for Retention
Chi-square + Cramér's V	Binary and categorical features	$p < 0.05$ AND Cramér's V > 0.05
Welch's T-test + Cohen's d	Continuous features	$p < 0.05$ AND |Cohen's d| > 0.2

Payment method was binary-encoded into four flags (payment_is_bacs, payment_is_card, payment_is_worldpay, payment_unknown) before the final feature set was assembled.

The final df_model.csv was built by combining features from three selection pools: hypothesis testing results, NZV lift analysis, and EDA gradient analysis - with leakage columns removed as a final safety check.

5. Feature Selection

Input: df_model.csv. Output: dataset/feature_store/final_pre-renewal_churn_dataset.csv

An 80/20 stratified train/test split was used for importance evaluation. Categorical columns were label-encoded with unseen test values mapped to UNKNOWN.

5.1 Tree-Based Importance

Model	Importance Metric	Settings
Random Forest	Mean decrease in impurity (Gini)	n_estimators=300, max_depth=8, class_weight=balanced
XGBoost	Total gain across splits	n_estimators=300, max_depth=6, scale_pos_weight=imbalance ratio

5.2 Combined Rank Aggregation

Rankings from both models were averaged to produce a combined rank:

```
avg_rank = (rf_rank + xgb_rank) / 2
```

The top 52 features by average rank formed the tree-selected set. This approach avoids selecting features that are important to only one model's inductive bias.

5.3 Business-Critical Feature Override

31 domain-validated features were force-added to the final set regardless of tree rankings, ensuring the model includes interpretable, business-meaningful predictors.

Category	Features
Tenure & Membership	Tenure_Years, Band, Last_Band, Current_Auto_Renewal_Flag
Pricing	Gross, Amount, price_increase_pct, price_change
Renewal Scores	Total_Renewal_Score_New, Renewal_Score_At_Release, Anchoring_Score, Sustainability_Score, Auto_Renewal_Score

Connections	#_of_Connection, Connection_Group, Anchor_Group
Call Signals	rc_Membership_Renewal_Decision, rc_desire_cancel_desired_to_cancel_true, rc_churn_cat_cost___price___too_expensive_true
CRM Signals	crm_contractor_suggested_leave, crm_customer_complained, crm_financial_hardship_mentioned, crm_agent_chase_count, crm_contractor_sentiment_score
Interaction Flags	had_cc_interaction, had_em_interaction, had_rc_interaction

Final feature count: tree-selected (52) + force-added business features not already in top 52. The dataset was saved with identifier columns (year_key, Co_Ref, Renewal_Year), features, and churn_label.

6. Model Building & Hyperparameter Tuning

Input: final_pre-renewal_churn_dataset.csv

6.1 Train / Test Split

A time-based split was used to simulate real-world deployment and prevent leakage:

Set	Criterion	Purpose
Train	Renewal_Year < 2025	Historical data - model learns from past
Test	Renewal_Year >= 2025	2025 renewals - unseen future data

Identifier columns (Co_Ref, year_key, Renewal_Year) were dropped from features but retained separately. Class imbalance ratio (majority / minority) was computed and used as scale_pos_weight for XGBoost.

6.2 Preprocessing for Modelling

Step	Method	Detail
Label encoding	sklearn LabelEncoder	Unseen test values mapped to −1
Numerical imputation	SimpleImputer(strategy='mean')	Fit on train, applied to both train and test
Final leakage drop	Manual	payment_timeframe_missing (target copy), payment_unknown (100% churn), crm_contractor_suggested_leave (zero variance in train), crm_contractor_sentiment_score (zero variance in train)

6.3 Leakage Checks

Three systematic checks were run before training: (1) Group-level churn rate check - flagged any value covering >5% of data with 100%/0% churn rate in low-cardinality columns. (2) Perfect separator check - flagged values in >1% of data with perfect class separation. (3) LightGBM quick feature importance scan - ensured no single feature dominated with a suspiciously large importance score. A train vs test PR-AUC gap check confirmed absence of severe overfitting.

6.4 Stage 1 - Baseline Models

Model	Class Balancing	Notes
Logistic Regression	class_weight='balanced'	Wrapped in StandardScaler pipeline (LR needs scaling)
Random Forest	class_weight='balanced'	n_estimators=300, max_depth=8
XGBoost	scale_pos_weight	n_estimators=300, max_depth=6, eval_metric='aucpr'
LightGBM	class_weight='balanced'	n_estimators=300
CatBoost	auto_class_weights='Balanced'	iterations=300

Models were ranked by PR-AUC. The top 2 were selected for hyperparameter tuning.

6.5 Stage 2 - Hyperparameter Tuning (Optuna)

Bayesian optimisation with 20 trials per model. Validation: 5-fold Stratified Cross-Validation on training set, maximising mean PR-AUC.

Model	Key Search Space
XGBoost	n_estimators: 200-600, max_depth: 4-10, learning_rate: 0.01-0.30 (log), subsample: 0.6-1.0, colsample_bytree: 0.6-1.0, scale_pos_weight: 6-12
LightGBM	n_estimators: 200-600, num_leaves: 20-150, max_depth: 4-12, learning_rate: 0.01-0.30 (log), feature_fraction: 0.6-1.0, bagging_fraction: 0.6-1.0
CatBoost	iterations: 200-600, depth: 4-10, learning_rate: 0.01-0.30 (log), l2_leaf_reg: 1-10

Threshold Integration: To ensure fair comparison, an **optimal classification threshold** was calculated for each trial by maximizing the F1-Score on the validation fold (searching 0.10 to 0.90, step 0.01). This allows hyperparameter selection to be based on the model's best achievable F1-performance rather than a default 0.5 boundary.

6.6 Stage 3 - Ensemble Methods

Ensemble	Method	Members
Stacking	Meta-learner (Logistic Regression) trained on base model predictions	Same members; base predictions via 5-fold CV to prevent meta-level leakage

The best model across all stages by test PR-AUC was selected as the final model.

Threshold Optimization: For the meta-learner, the final decision threshold was optimized by maximizing a F1-Score This reflects the business cost where missing a churner (False Negative) is more detrimental than incorrectly flagging a retained customer (False Positive). The optimal threshold was selected from a range of 0.10 to 0.90.

6.8 Final Evaluation

Metric	Purpose
PR-AUC	Primary - overall model quality for the minority churn class
ROC-AUC	Secondary - overall discrimination ability
Recall	Churn capture rate - what fraction of churners were caught
Precision	Flag accuracy - what fraction of flagged accounts actually churned
F1-Score	Harmonic mean of precision and recall

Outputs: confusion matrix, business-readable summary (TP/FN/FP/TN with churn capture rate and flag accuracy), and Precision-Recall curve with operating point marked at the optimal threshold.

7. Key Design Decisions

Decision	Rationale
PR-AUC as primary metric	Class imbalance makes accuracy and ROC-AUC misleading; PR-AUC focuses on minority class performance
F1-Score for threshold tuning	Used for threshold tuning to balance the trade-off between precision and recall, specifically weighted to prioritize the detection of churners over the cost of false alarms.
Time-based train/test split	Prevents leakage and simulates deployment: train on past, predict future
Leave outliers as-is	Tree-based models are robust to outliers; transformations would remove meaningful business signals
Lift-based NZV retention	Rare events with 2× churn lift carry genuine predictive power despite low frequency
Tri-state encoding	Distinguishes 'no call happened' from 'call happened but signal absent' - critical for interaction features
Business-critical feature override	Ensures interpretable, domain-validated features enter the model even if not top-ranked by tree importance
Domain-specific null fills	E.g., 0 for missing connections (no connections), current flag for proforma flags (reliable proxy) - more accurate than mean/median imputation

8. Glossary

Term	Definition
PR-AUC	Area under the Precision-Recall curve - primary model evaluation metric
ROC-AUC	Area under the Receiver Operating Characteristic curve
F1-Score	Harmonic mean of precision and recall, providing a balanced evaluation of model performance on the minority churn class.
Churn label	Binary target: 1 = churned (did not renew), 0 = won (renewed)
year_key	Composite join key: Co_Ref + '_' + Renewal_Year
Pre-renewal window	14 days before Prospect_Renewal_Date - period from which CC/renewal call data is used
Tri-state encoding	Splitting No_Interaction / False / True into two binary columns (_interacted and _true)
scale_pos_weight	XGBoost parameter: ratio of majority to minority class, used to handle imbalance
NZV	Near-zero variance - columns where one value appears in ≥99% of rows
Lift	Churn rate in a segment divided by overall churn rate - measures how much a feature raises churn probability
Optuna	Bayesian hyperparameter optimisation framework using successive halving and TPE sampler
Stacking	Ensemble where base model predictions are used as features for a meta-learner
Soft Voting	Ensemble where final probability is the average of all base model probabilities